Community detection in geometric graphs: Recent advances until 2026

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Résumé TODO : introduction UGCM, presenter la litterature

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1 Introduction

Community detection consists in finding clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, etc. More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020] word embedding, physic, ...

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. This model is now well understood, see [Abbe, 2018]. Nonetheless, it has some drawbacks, since it is unrealistic to think that every node can connect to every other one with the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Adding geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003] and more recently [Duchemin and De Castro, 2023]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels of inference. First, *distinguishability* asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

— *Weak/partial recovery* : detecting communities better than random guessing
— *Almost exact recovery* : the fraction of misclassified vertices vanishes as the number of nodes grows.
— *Exact recovery* : all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. A related but distinct line of work considers mixture based latent position models, such as the Gaussian Mixture Block Model [Li and Schramm, 2024], where community labels determine node positions using gaussian distributions. See [Reynolds et al., 2009] [Rasmussen, 1999]. This differs from the geometric models considered here, where geometry directly affects edge formation rather than determining node placement. We therefore don't include these models in the core of this survey.

Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label).

2 Problem setting and inference tasks

Formally, we observe a graph $G := (V, E)$, where V is the vertex set and E is the set of edges between vertices. Throughout the report, n denotes the asymptotic scaling parameter. Depending on the model, the observed number of vertices may be either deterministic or random, but is of order n. We consider a simple undirected graph Une remarque là dessus ? Pourquoi non-orienté est classique comme hypothese ?, without self-loops. The vertices are partitioned into k communities $C_1, \dots, C_k$, so $V = \bigsqcup_{i=1}^{k} C_i$. We denote by $\sigma^\star : V \to [k]$ the ground-truth community label function, defined by $\sigma^\star(v) = i$ whenever $v \in C_i$. The community structure is hidden, and our goal is to recover it from the observed data.

An estimator is therefore a function of the observations. Depending on the model, the observations may consist only of the graph G, or of the graph together with the node locations $X = (X_v)_{v \in V}$. Accordingly, we write the estimator as $\hat{\sigma} = \hat{\sigma}(G)$ or $\hat{\sigma} = \hat{\sigma}(G, X)$, and we view $\hat{\sigma} : V \to [k]$ as an estimated labeling. In general, the labels are only identifiable up to a relabeling of the communities. Indeed, permuting the names of the communities does not change the underlying partition of the vertex set. For this reason, the quality of an estimator is measured up to permutation. Denoting by $\mathfrak{S}_k$ the symmetric group on $[k]$, we define

$$A_n(\hat{\sigma}, \sigma^\star) \coloneqq \frac{1}{|V|} \max_{\pi \in \mathfrak{S}_k} \sum_{u \in V} \mathbf{1}\big\{\hat{\sigma}(u) = \pi(\sigma^\star(u))\big\}$$

Thus, $A_n(\hat{\sigma}, \sigma^\star)$ measures the fraction of correctly classified vertices after the best possible relabeling of the communities. We now give more formal definitions of the problems introduced in Section 1.

Distinguishability. Following [Sankararaman and Baccelli, 2018], distinguishability is a hypothesis testing problem. Let $(\mathbb{P}_n)_{n \geq 1}$ be the laws of a geometric random graph model with community structure and let $(\mathbb{Q}_n)_{n \geq 1}$ be the laws of a null geometric model without communities, both defined on the same observation space. Let Ω_n be the observation space. A test is a measurable function $\varphi_n : \Omega_n \to \{0, 1\}$ where $\varphi_n = 1$ means that we decide in favor of the community model. We say that $(\mathbb{P}_n)$ and $(\mathbb{Q}_n)$ are distinguishable if there exist tests $(\varphi_n)_{n \geq 1}$ and a constant $\gamma > 0$ such that, under the uniform prior on $\{\mathbb{P}_n, \mathbb{Q}_n\}$,

$$\mathbb{P}\big(\varphi_n \text{ makes the correct decision}\big) = \frac{1}{2}\,\mathbb{P}_n(\varphi_n = 1) + \frac{1}{2}\,\mathbb{Q}_n(\varphi_n = 0) \geq \frac{1}{2} + \gamma - o_n(1)$$

Weak/Partial recovery. We say that weak/partial recovery is achievable if the estimator performs better than random guessing, that's to say in the symmetric case, if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ and a constant $\alpha > 1/k$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq \alpha\big) \xrightarrow[n \to \infty]{} 1$$

Almost exact recovery. We say that almost exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that for every $\varepsilon > 0$,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq 1 - \varepsilon\big) \xrightarrow[n \to \infty]{} 1$$

Exact recovery. We say that exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n\geq 1}$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) = 1\big) \xrightarrow[n\to\infty]{} 1$$

Most works in the literature focus on the case $k = 2$. Although several ideas extend beyond this setting, this is also the framework that we adopt in the following, except when specified otherwise. In that case, it is convenient to encode the labels by $\{-1, +1\}$, instead of $[2] = \{1, 2\}$. Then "up to permutation" reduces to a global sign flip.

Each vertex also belongs to a space $\mathcal{X}$. In the following, we specify each time whether this space is latent or observed. For theoretical analysis, $\mathcal{X}$ is often chosen for mathematical convenience. The Euclidean space $\mathbb{R}^{d-1}$ is a natural reference because of its symmetry, homogeneity, isotropy, and direct geometric interpretation. See [Abbe et al., 2021], Citer ceux qui sont euclidiens. In the following, d is an integer. We also encounter the torus, mainly to avoid boundary effects and to focus on the core ideas of the proofs, Todo citer. Lastly, for latent models, the high-dimensional unit sphere $\mathbb{S}^{d-1}$ is a standard reference space to model similarity, and it also provides convenient spectral properties [Castro et al., 2020]. A summary of theoretical works on $\mathbb{S}^{d-1}$ is given in [Duchemin and De Castro, 2023].

In Section 3 we present models used throughout the literature. Hence, we suggest a model to unify them while remaining close to the notation used in the papers. We call it the *Unified Geometric Community Model* (UGCM) and it is defined as,

$$\mathrm{UGCM}\,(\mathcal{X}, \rho, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

where $\mathcal{X}$ is the ambient space and ρ is its metric. $W_n \subset \mathcal{X}$ is the window where nodes are sampled at the scale n. μ_n governs the spatial sampling mechanism. Usually, the vertices are sampled either as n i.i.d. points in W_n with law μ_n or as a Poisson Point Process (PPP) on W_n with intensity measure μ_n. We denote by Z the label set, and by $\pi = (\pi_i)_{i\in Z}$ the distribution of the labels. For all $u \in V$ and independently, $\sigma^\star(u) \sim \pi$. $\mathcal{Y}$ the space of pairwise observation. Let $\mathcal{P}(\mathcal{Y})$ denote the set of probability distributions on $\mathcal{Y}$, we consider the kernel observation,

$$K_n : \mathcal{X} \times \mathcal{X} \times Z \times Z \to \mathcal{P}(\mathcal{Y})$$

and for each unordered pair of vertices u, v, we observe,

$$Y_{uv} \mid (X, \sigma^\star) \sim K_n\big(X_u, X_v, \sigma^\star(u), \sigma^\star(v)\big)$$

The kernel is the core of the model. We note that Soft-RGG introduced by [Penrose, 2016] and developed in [Duchemin and De Castro, 2023] can be seen as a special case of UGCM where there exists a function $f_n : \mathbb{R}^+ \to [0, 1]$ such that for every $x, y \in \mathcal{X}$ and $z, z' \in Z$, $K_n(x, y, z, z') = \mathrm{Ber}(f_n(\rho(x, y)))$. Moreover, a particular case of Soft-RGG are the usual random geometric graph, i.e. the hard-threshold special case, when f_n is an indicator function, i.e. there exists $r_n > 0$ such that $K_n(x, y, z, z') = \delta_{\mathbf{1}\{\rho(x,y) \le r_n\}}$. Then, this framework also accommodates weighted graphs, since the observation space $\mathcal{Y}$ is not necessarily binary.

As we will see, several phase-transition phenomena depend on the connectivity scale of the graph, which is usually measured through the average degree. Let D_n denote the degree of a vertex chosen uniformly at random. We can distinguish three regimes of connectivity, depending on the growth of $\mathbb{E}(D_n)$ as $n \to \infty$. The most realistic but also the most delicate regime is the *sparse regime*. Each vertex only carries a finite amount of local information, the expected degree remains bounded,

$$\mathbb{E}(D_n) = \Theta(1)$$

In the *logarithmic regime*, the graph becomes sufficiently connected for exact recovery to become meaningful,

$$\mathbb{E}(D_n) = \Theta(\log n)$$

Finally, the *dense regime* corresponds to the easiest setting, the graph carries a lot of information,

$$\mathbb{E}(D_n) = \Theta(n)$$

3 Geometric community models and recovery thresholds

3.1 Geometric Block Model (GBM)

Historically, one of the earliest geometric models for community detection is the Geometric Block Model (GBM) introduced in [Galhotra et al., 2018]. They mainly study the case of two symmetric communities. In its general form, the model places each vertex at a latent position on the unit sphere $\mathcal{X} := \mathbb{S}^{d-1}$, sampled uniformly, and edges are drawn through a hard geometric rule. For $d \in \mathbb{N} \backslash \{0, 1\}$, $r_{in}, r_{out} \in [-1, 1]$, $r_{ij} := r_{in}\delta_{ij} + r_{out}(1 - \delta_{ij})$, they define

$$\mathrm{GBM}(r_{in}, r_{out}, d) := \mathrm{UGCM}(\mathcal{X}, \|\cdot\|_2,\ \mathcal{X}, \mathrm{Unif}(\mathcal{X}), \{-1, +1\}, \mathrm{Ber}(1/2), \{0, 1\}, K_n)$$

with $K_n : (x, y, z, z') \mapsto \mathbf{1}\{\langle x, y\rangle \ge r_{z,z'}\}$. The condition $r_{in} > r_{out}$ ensures that edges within communities are denser than across communities. The space $\mathcal{X}$ is assumed latent, so recovery is done only from the adjacency matrix.

They study the one dimensional circular case, corresponding to $\mathbb{S}^1 \subset \mathbb{R}^2$, hence $d = 2$ in the notation above, in the regime

$$r_{in} := a\frac{\log n}{n} \qquad r_{out} := b\frac{\log n}{n} \qquad a > b > 0$$

This corresponds to the logarithmic regime. This regime matches the connectivity scale of one dimensional random geometric graphs. Their main algorithm is motif based, that is, for an observed edge $(u, v) \in E$, they count the number of common neighbors of u and v, so the number of triangles containing this edge. Then they use this statistic to decide whether u and v belong to the same community. The geometry creates strong local dependence and transitivity. Unlike in the SBM, triangles are informative in the GBM. Still in this regime, the simple motif counting procedure already yields exact recovery with high probability when $r_{in} \geq 4r_{out}$. The algorithm is efficient, and the journal version later clarifies that this one threshold guarantee from the conference paper is order optimal up to constants but restricted to that parameter range [Galhotra et al., 2018, 2023].

Firstly, [Galhotra et al., 2020] introduces the random annulus graph (RAG). Let $I := \left[b \left(\frac{\log n}{n} \right)^{1/d}, a \left(\frac{\log n}{n} \right)^{1/d} \right]$, they work on $\mathbb{S}^d$ with $RAG_d(n, I)$, a RGG where nodes are connected only if the distance between them falls inside I, hence the name annulus graph. In dimension 1, they obtain a sharp connectivity threshold for RAG : the graph is connected with high probability if $a > 1$ and $a - b > 1/2$, and disconnected with high probability if $a < 1$ or $a - b < 1/2$. For $d > 1$, they give only sufficient conditions [Galhotra et al., 2020].

Secondly, the main algorithmic refinement is to replace the original one threshold triangle rule by a two threshold procedure. The reason is that, once one leaves the regime $r_{in} \geq 4r_{out}$, a single threshold is no longer sufficient. For inter community edges, the normalized triangle count is concentrated around $2r_{out}$, whereas for intra community edges it depends on the latent distance and may lie either above or below that value. They therefore define two thresholds

$$E_S := (2b + f_1)\frac{\log n}{n} \qquad E_D := (2b - f_2)\frac{\log n}{n}$$

where f_1 and f_2 are chosen from Chernoff type large deviation inequalities. Edges whose triangle count falls inside the ambiguous interval $[E_D, E_S]$ are removed. This deletes all inter community edges with high probability, while preserving enough informative intra community edges. It remains to show that the surviving intra community edges still connect each community, and this is exactly where the RAG connectivity results are used.

As a consequence, they obtain exact recovery in dimension 1 with explicit conditions and an impossibility result when $a < 1$ or $a - b < 1/2$. However, the exact recovery threshold for the latent one dimensional GBM is still not fully characterized, see Figure 1. A useful contrast is that this gap disappears when the latent positions are revealed. In that case, the one dimensional GBM has a sharp exact recovery threshold which coincides with the sharp connectivity threshold of the RAG. In higher dimension $d > 1$, they also prove a polynomial time exact recovery result at the correct order of magnitude,

$$r_{out} = \Theta\left(\left(\frac{\log n}{n} \right)^{1/d} \right) \quad \text{and} \quad r_{in} - r_{out} = \Theta\left(\left(\frac{\log n}{n} \right)^{1/d} \right)$$

are sufficient, while recovery becomes impossible if one of these quantities is of smaller order [Galhotra et al., 2023].

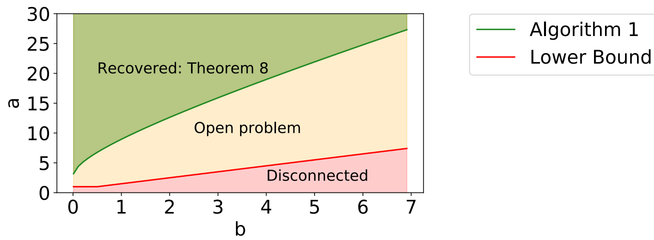

FIGURE 1. Phase diagram for exact recovery in the one dimensional latent GBM in the regime $r_{in} = a \log n/n$ and $r_{out} = b \log n/n$, reproduced from Galhotra et al. [2023], Figure 5. The red region corresponds to disconnected graphs, hence an impossibility result. The yellow region remains an open problem, and the green region corresponds to the parameter range certified by Algorithm 1 (Theorem 8).

Finally, [Chien et al., 2020] studies active learning in the latent GBM with two communities. The model is the same, but the learner is now allowed to query the labels of a small number of carefully chosen vertices to a perfect oracle. Their approach keeps the motif counting idea as a first step, in order to remove many inter community edges, and then uses label queries in a second step. They propose two algorithms. The first combines this pruning step with the graph based active learning procedure S^2 [Dasarathy et al., 2015], while the second is more aggressive and then queries one label per connected component. Their main result is that exact recovery becomes possible with high probability using only a sublinear number of queries, even in parameter regimes where the best known unsupervised guarantees fail. This gives a smooth trade off between the hardness of clustering and the query complexity. They do not prove any optimality result for this active setting. Moreover, the theoretical bounds seem somewhat pessimistic. Although the analysis favors the first algorithm, the paper notes that the second often uses fewer queries in practice, which suggests that the proof can likely be improved [Chien et al., 2020].

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] introduce the Geometric Stochastic Block Model (GSBM) which is later developed in full detail in [Abbe et al., 2021]. The geometric positions are observed, the space $\mathcal{X} := \mathbb{R}^d$ is not latent. Here, n is a scaling parameter and not the number of nodes. Points are sampled as a PPP on a window $W_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$ with intensity measure $\mu_n := \lambda \mathrm{Leb}_{|W_n}$ with $\lambda > 0$. The window W_n has a volume n. They study the case of two symmetric communities. Nodes are connected according to their distance and labels through $f_{in}^{(n)}$ and $f_{out}^{(n)}$. We assume for all $r \geq 0$, $1 \geq f_{in}^{(n)}(r) \geq f_{out}^{(n)}(r) \geq 0$ and $f_{in}^{(n)} \neq f_{out}^{(n)}$ up to Lebesgue measure 0. More assumptions on these functions are given according to the regime that we are interested. The model is then defined as,

$$\mathrm{GSBM}(f_{in}^{(n)}, f_{out}^{(n)}, \lambda, d) \coloneqq \mathrm{UGCM}(\mathcal{X}, \rho_n, W_n, \mu_n, \{-1,+1\}, \mathrm{Ber}(1/2), \{0,1\}, K_n)$$

where the metric is the Euclidean distance in the sparse regime, $\rho_n \coloneqq \|\cdot\|_2$, and toroidal metric in the logarithmic regime, $\rho_n \coloneqq \|\cdot\|_{T_n}$. The latter is to avoid boundary effects on W_n due to the infinity of neighbors a points can have, for the sparse regime this consideration is negligible. Moreover, the kernel is defined as,

$$K_n : (x,y,z,z') \longmapsto \mathrm{Ber}\left(f_{in}^{(n)}(\rho_n(x,y))\mathbf{1}\{z=z'\} + f_{out}^{(n)}(\rho_n(x,y))\mathbf{1}\{z\neq z'\}\right)$$

In the sparse regime, nodes have a finite expected degree. Accordingly, we assume that the connection functions do not depend on n, namely, $f_{in}^{(n)} = f_{in}$ and $f_{out}^{(n)} = f_{out}$, and that

$$\int_{\mathbb{R}^d} f_{in}(\|x\|_2) + f_{out}(\|x\|_2)\mathrm{d}x < \infty$$

The metric is Euclidean, which allows the finite graphs to be viewed as restrictions of a single infinite random graph [Last and Penrose, 2018]. In this regime, the main question is weak recovery. By monotonicity, there exists a critical value $\lambda_c \in [0,\infty]$ separating the unsolvable and solvable phases. The first result is an impossibility criterion based on the auxiliary random connection model $H_{\lambda,g(\cdot),d}$ defined on the same PPP of intensity λ where two points at distance r are connected independently with probability $g(r)$. Denoting by $\theta(H_{\lambda,g(\cdot),d})$ the percolation probability, the probability that the connected component of a typical node is infinite. They prove that if $\theta(H_{\lambda,f_{in}-f_{out},d}) = 0$ then weak recovery is impossible. In other words, if the informative part $f_{in} - f_{out}$ does not generate an infinite connected component, then the community information cannot propagate at long range. In particular, for $d=1$, weak recovery is impossible for every $\lambda > 0$. To prove the lower bound, they introduce an auxiliary problem called Information Flow from Infinity. They prove that this problem is easier than community detection, so its impossibility already implies the impossibility of weak recovery [Abbe et al., 2021]. For $d \geq 2$, this yields the explicit lower bound,

$$\lambda \leq \left(\int_{\mathbb{R}^d} \big(f_{in}(\|x\|_2) - f_{out}(\|x\|_2)\big)\, dx\right)^{-1} \quad \Longrightarrow \quad \text{weak recovery is impossible}$$

On the positive side, they introduce the Good Bad Grid (GBG) algorithm, its complexity is $O(n^2)$. Although the graph is globally sparse, it remains locally dense because geometry creates many short loops and common neighbors. Hence, for two nearby nodes, the number of common neighbors carries information on whether they belong to the same community. The algorithm first compares nearby pairs using such local statistics, then partitions the space into small cells and propagates consistent decisions across the grid. Its analysis relies on a coupling with a dependent site percolation process. They prove that if $d \geq 2$ then there exists $\lambda_{upper} < \infty$ such that weak recovery is solvable for all $\lambda \geq \lambda_{upper}$. Combined with the lower bound above, this implies $0 < \lambda_c < \infty$ so the sparse regime exhibits a non-trivial phase transition [Abbe et al., 2021]. In some special cases, they obtain a sharp characterization of the threshold. If connection functions are hard thresholds, that is, if there exist $R_{in}, R_{out} \geq 0$ such that,

$$f_{in} : r \mapsto \mathbf{1}_{\{r \leq R_{in}\}} \qquad \text{and} \qquad f_{out} : r \mapsto \mathbf{1}_{\{r \leq R_{out}\}} \qquad \text{and} \qquad R_{in} \geq R_{out}$$

then weak recovery is solvable if and only if $\theta(H_{\lambda, f_{in} - f_{out}, d}) > 0$. So, in this hard threshold case, the phase transition is fully characterized. More generally, they also show a weak form of asymptotic optimality for GBG, when the signal is strong enough, the fraction of correctly classified vertices can be made arbitrarily close to 1. In our notation, this means that for every $\varepsilon > 0$, there exists $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$,

$$\lim_{n \to \infty} \mathbb{P}\big(A_n(\hat{\sigma}_n^{GBG,\lambda}, \sigma^\star) \geq 1 - \varepsilon\big) = 1$$

A related question, well studied in the SBM, is whether a graph with communities can be distinguished from a null model without community structure. In [Abbe et al., 2021], the authors formulate the same testing problem for the GSBM. The null model is a random connection model without communities, with the same spatial distribution of nodes and connection function $(f_{in} + f_{out})/2$. Hence, the null and alternative models have the same average degree. They prove that, as soon as $f_{in} \neq f_{out}$ on a set of positive Lebesgue measure, the two induced probability measures are mutually singular. Therefore, the distinguishability problem is solvable with probability $1 - o_n(1)$ for every $\lambda > 0$, so it exhibits no phase transition. They construct an explicit test based on local triangle profiles. An interesting fact is that, in the sparse regime, one may detect the presence of communities while still being unable to recover a partition better than random. This is in contrast with the sparse SBM with two communities, where distinguishability and weak recovery coincide [Mossel et al., 2012].

In the logarithmic regime, the connection functions now depend on n and satisfy for $C_{in} > C_{out} \geq 0$,

$$\int_{\mathbb{R}^d} f_{in}^{(n)}(\|x\|_{T_n})\, dx = C_{in} \log n \qquad \int_{\mathbb{R}^d} f_{out}^{(n)}(\|x\|_{T_n})\, dx = C_{out} \log n$$

This is also the natural scale of connectivity for geometric graphs, which explains why exact recovery is studied in this regime. However, connectivity is not a necessary condition for exact recovery in the GSBM since the node locations are observed, isolated or weakly connected vertices may still be classified from their geometric position relative to the others. For the impossibility result, they introduce the genie estimator and the notion of a flip bad vertex, namely a vertex whose label is misclassified by the local likelihood test when all other labels are revealed. Writing N_n^{flip} for the number of such vertices, they prove that if $\mathbb{E}(N_n^{\text{flip}}) \xrightarrow[n\to\infty]{} \infty$ and if long range correlations between flip bad events are negligible, then $N_n^{\text{flip}} > 0$ with high probability. Therefore exact recovery is impossible [Abbe et al., 2021]. On the positive side, they adapt the GBG algorithm to this regime. The local pairwise tests become much more accurate, since the number of common neighbors is now of order $\log n$, and the proof no longer relies on the existence of a giant good component. Instead, they show that when the signal is strong enough, every cell is good with high probability, which yields exact recovery [Abbe et al., 2021].

They then focus on a particular logarithmic model, defined by

$$f_{in}^{(n)} : r \mapsto a\,\mathbf{1}_{\{r\le(\log n)^{1/d}\}} \qquad f_{out}^{(n)} : r \mapsto b\,\mathbf{1}_{\{r\le(\log n)^{1/d}\}} \qquad 0 \le b < a \le 1$$

This should be viewed as a non trivial instance of the logarithmic regime. Denote by ν_d the volume of the unit ball in $\mathbb{R}^d$. In this case, they obtain an explicit necessary condition for exact recovery,

$$\lambda\nu_d\Big(1-\sqrt{ab}-\sqrt{(1-a)(1-b)}\Big) < 1 \quad \Longrightarrow \quad \text{exact recovery is impossible} \tag{1}$$

Then, they also prove that there exists $C(a,b,d) \ge 0$ such that $\lambda\nu_d\Big(1-\sqrt{ab}-\sqrt{(1-a)(1-b)}\Big) > C(a,b,d)$ is sufficient for exact recovery by GBG. They conjecture that the lower bound is in fact sharp. This conjecture was later confirmed by [Gaudio et al., 2024a].

More recently, [Gaudio and Jin, 2025] extended this sharp result. They consider a model where edges are only possible if they are in a visibility radius $r(\log n)^{1/d}$ with $r > 0$, and assume mild regularity conditions on the connection functions such as supported on $[0,r]$, a finite number of intersections between f_{in} and f_{out}, and no interval on which the two functions remain arbitrarily close. This includes in particular the previous indicator case, but also broad classes of piecewise regular or continuous kernels. Their main result is a complete characterization of the exact recovery threshold through the information quantity

$$I(f_{in}, f_{out}) := \lambda\nu_d r^d \int_0^r \Big(1-\sqrt{f_{in}(t)f_{out}(t)}-\sqrt{(1-f_{in}(t))(1-f_{out}(t))}\Big)\frac{d\,t^{d-1}}{r^d}\,dt$$

which is a generalized Chernoff Hellinger divergence. They prove that exact recovery is achievable when $I(f_{in}, f_{out}) > 1$, while it is impossible when $I(f_{in}, f_{out}) < 1$. In dimension $d = 1$, one must additionally have $\lambda r > 1$, and exact recovery is impossible when $\lambda r < 1$. Hence, the paper shows that the exact recovery problem in the two community GSBM admits a sharp information theoretic threshold far beyond the step function case. They also provide a polynomial time algorithm achieving this threshold.

3.3 Geometric Kernel Block Model (GKBM)

[Avrachenkov et al., 2024] introduce the Geometric Kernel Block Model (GKBM). The geometric positions are observed, so the space $\mathcal{X} := \left]-\frac{1}{2}, \frac{1}{2}\right]$

is not latent, and it is equipped with the toroidal metric ρ. Here again, n is a scaling parameter and not the number of nodes. Points are sampled as a PPP on $\mathcal{X}$ with intensity measure $\mu_n := \lambda n \mathrm{Leb}_{|\mathcal{X}}$ with $\lambda > 0$. They focus in the one dimensional case with two symmetric communities, although the techniques may extend to higher dimensions. The link probabilities are defined according to $\phi : \mathbb{R}^+ \to [0,1]$ measurable with a support bounded by $r > 0$. Let $a, b \in [0,1]$,

$$\mathrm{GKBM}(\lambda, a, b, \phi) := \mathrm{UGCM}(\mathcal{X}, \rho, \mathcal{X}, \mu_n, \{-1,+1\}, \mathrm{Ber}(1/2), \{0,1\}, K_n)$$

Moreover they study the logarithmic regime so the kernel is set as,

$$K_n : (x, y, z, z') \longmapsto \mathrm{Ber}\Bigg(\big(a\mathbf{1}\{z = z'\} + b\mathbf{1}\{z \neq z'\}\big)\phi\Big(\frac{n}{\log n}\rho(x,y)\Big)\Bigg)$$

The hard threshold version of the logarithmic GSBM leads to the explicit bound (1). For the GKBM, the authors replace this hard threshold by a more general kernel on their restricted one dimensional model, and introduce the information quantity

$$I_\phi(a,b) := 2\int_{\mathbb{R}_+} \left(1 - \sqrt{ab}\,\phi(x) - \sqrt{(1 - a\phi(x))(1 - b\phi(x))}\right) dx$$

This quantity plays the same role as in [Abbe et al., 2021]. In fact, when $\phi = \mathbf{1}_{[0,1]}$, it reduces to the one dimensional hard threshold case. Their main result shows that exact recovery is impossible if $\lambda r < 1$ or $\lambda I_\phi(a,b) < 1$. Conversely, if $\lambda r > 1$ and $\lambda I_\phi(a,b) > 1$, then exact recovery is achievable under the additional assumption that $\phi(x) > 0$ for all $x \in [0,r]$ [Avrachenkov et al., 2024]. They also propose a two phase algorithm, adapted from the recent GSBM exact recovery approach, where labels are first recovered on a small block and then propagated. The paper states that this procedure runs in time linear in the number of edges, this corresponds to a complexity of $O(n \log n)$, in the logarithmic regime [Avrachenkov et al., 2024].

3.4 Geometric Hidden Community Model (GHCM)

The last model known to us is the Geometric Hidden Community Model (GHCM), as formulated in [Gaudio and Jin, 2026]. The geometric positions are observed, so the space $\mathcal{X} := \mathbb{R}^d$ is not latent. Let $W_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$ equipped with the toroidal metric ρ_n. Let $k \in \mathbb{N}$, Z be a label set with $|Z| = k$ and prior $\pi = (\pi_i)_{i \in Z}$. Points are sampled as a PPP on W_n with intensity measure $\mu_n := \lambda \mathrm{Leb}_{|W_n}$ with $\lambda > 0$. The big improvement is to consider weighted graphs, hence edges are not solely 0 or 1 but any real value, $\mathcal{Y} := \mathbb{R}$. The most general GHCM model, from [Gaudio and Jin, 2026], is distance dependent and is parametrized by a family of pairwise observation laws. For some $r > 0$, let

$$P = \big(P_{zz'}(t)\big)_{z,z' \in Z,\ t \in [0,r]} \in \mathcal{P}(\mathcal{Y})^{Z \times Z \times [0,r]}$$

where $P_{zz'}(t)$ is the law of the observation between two vertices of labels z and z' at distance t. Since the graph is undirected, we assume $P_{zz'}(t) = P_{z'z}(t)$. In the papers, these laws are taken either discrete or with density with respect to the Lebesgue measure on $\mathbb{R}$. We write $p_{zz'}(\cdot;t)$ for a probability mass function or a density of $P_{zz'}(t)$.

$$\mathrm{GHCM}(P, \lambda, \pi, k, d, r) := \mathrm{UGCM}(\mathcal{X}, \rho_n, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

The kernel K_n is then given by

$$K_n : (x, y, z, z') \longmapsto \begin{cases} P_{zz'}\left(\dfrac{\rho_n(x,y)}{(\log n)^{1/d}}\right) & \text{if } \rho_n(x,y) \leq r(\log n)^{1/d} \\ \delta_0 & \text{otherwise} \end{cases}$$

Thus geometry determines whether an observation is available and changes the law of this observation through the distance. This model definition allows us to capture problem such as $\mathbb{Z}^2$-synchronisation, submatrix localization and more.

The original GHCM of [Gaudio et al., 2024a] is recovered when $P_{zz'}(t)$ does not depend on t, so geometry only acts as an observation mask, it only determines whether this law is sampled or not. Hence the visibility radius is fixed to $(\log n)^{1/d}$, which corresponds to $r := 1$. This model contains the GSBM as the Bernoulli special case. The paper studies exact recovery in the logarithmic regime and identifies the information theoretic threshold,

$$\lambda \nu_d \min_{z \neq z'} D_+(\theta_z, \theta_{z'}) = 1 \qquad \text{with} \qquad \theta_z := (p_{z1}, \dots, p_{zk})$$

with the additional one dimensional $d = 1$ obstruction, $\lambda < 1$. In the present discrete/continuous setting, D_+ is the Chernoff-Hellinger divergence given by

$$D_+(\theta_z, \theta_{z'}) = 1 - \inf_{t \in [0,1]} \sum_{r \in Z} \pi_r \int p_{zr}(y)^t p_{z'r}(y)^{1-t} \, dy$$

with the integral replaced by a sum in the discrete case. Above the threshold, exact recovery is achieved by a polynomial time algorithm in two phases under two main technical assumptions : a bounded log-likelihood ratio, and a distinctness condition requiring that for every $z \in Z$, the laws P_{zi} and P_{zj} are different whenever $i \neq j$. In dimension 1, when $|\Omega_{\pi,P}| = 1$, the proof uses the stronger assumption that all laws $P_{zz'}$ are pairwise distinct without needed $\lambda > 1$. Above the threshold, exact recovery is achieved by a polynomial time algorithm in two phases under two main technical assumptions : a bounded log-likelihood ratio, and a distinctness condition requiring that for every $z \in Z$, the laws P_{zi} and P_{zj} are different whenever $i \neq j$. In dimension 1, it requires the additional condition $\lambda > 1$. This extra assumption only appears when the model still has non trivial permissible relabelings, that is, when the set

$$\Omega_{\pi,P} := \{\omega : Z \to Z \text{ permutation}, \forall z, z' \in Z pi_z = \pi_{\omega(z)} \text{ and } P_{zz'} = P_{\omega(z)\omega(z')}\}$$

contains more than the identity. In the special case $|\Omega_{\pi,P}| = 1$, the paper instead uses the stronger assumption that all laws $P_{zz'}$ are pairwise distinct, and then no longer needs $\lambda > 1$.

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]

(papier SBM mais théorie intéressante) [Bordenave et al., 2015]

Cadre spectral intéressant pour certains espaces [Méliot, 2019]

Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]

Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geomemetric SBM) [Avrachenkov et al., 2021]

[Allem et al., 2025]

5 Ouvertures

— Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
  — Clustering [Krioukov, 2016]
  — Degre inhomogène [Barabási and Albert, 1999]
  — "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]

  Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours ! + definition d'un modele pour "régler" ça [Guarino et al., 2025]

  [Li et al., 2018]

  Nous devons estimer les paramètres des modèles. Dans le cas de Sankararamn et Baccelli, GBG peut être adapté.

Bibliographie

Emmanuel Abbe. Community detection and stochastic block models : recent developments. *Journal of Machine Learning Research*, 18(177) :1–86, 2018.

Emmanuel Abbe, Francois Baccelli, and Abishek Sankararaman. Community detection on euclidean random graphs. *Information and Inference : A Journal of the IMA*, 10(1) :109–160, 2021.

Luiz Emilio Allem, Konstantin Avrachenkov, Carlos Hoppen, Hariprasad Manjunath, and Lucas Siviero Sibemberg. Multi-community spectral clustering for geometric graphs, 2025. URL `https://arxiv.org/abs/2508.00893`.

Konstantin Avrachenkov, Andrei Bobu, and Maximilien Dreveton. Higher-order spectral clustering for geometric graphs. *Journal of Fourier Analysis and Applications*, 27(2) :22, 2021.

Konstantin Avrachenkov, BR Kumar, and Lasse Leskelä. Community detection on block models with geometric kernels. *arXiv preprint arXiv :2403.02802*, 2024.

Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439) :509–512, October 1999. ISSN 1095-9203. `https://doi.org/10.1126/science.286.5439.509`. URL `http://dx.doi.org/10.1126/science.286.5439.509`.

Charles Bordenave, Marc Lelarge, and Laurent Massoulié. Non-backtracking spectrum of random graphs : community detection and non-regular ramanujan graphs. In *2015 IEEE 56th Annual Symposium on Foundations of Computer Science*, pages 1347–1357. IEEE, 2015.

Matteo Bruno, Sandro Ferreira Sousa, Furkan Gursoy, Matteo Serafino, Francesca V Vianello, Ana Vranić, and Marián Boguñá. Community detection in the hyperbolic space. *arXiv preprint arXiv :1906.09082*, 2019.

Sébastien Bubeck, Jian Ding, Ronen Eldan, and Miklós Z Rácz. Testing for high-dimensional geometry in random graphs. *Random Structures & Algorithms*, 49(3) :503–532, 2016.

Yohann De Castro, Claire Lacour, and Thanh Mai Pham Ngoc. Adaptive estimation of nonparametric geometric graphs, 2020. URL `https://arxiv.org/abs/1708.02107`.

Yuxin Chen, Govinda Kamath, Changho Suh, and David Tse. Community recovery in graphs with locality. In *International conference on machine learning*, pages 689–698. PMLR, 2016.

Eli Chien, Antonia Tulino, and Jaime Llorca. Active learning in the geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 3641–3648, 2020.

Gautam Dasarathy, Robert Nowak, and Xiaojin Zhu. S2 : An efficient graph based active learning algorithm with application to nonparametric classification, 2015. URL `https://arxiv.org/abs/1506.08760`.

Charo I. Del Genio, Thilo Gross, and Kevin E. Bassler. All scale-free networks are sparse. *Physical Review Letters*, 107(17), October 2011. ISSN 1079-

7114. `https://doi.org/10.1103/physrevlett.107.178701`. URL `http://dx.doi.org/10.1103/PhysRevLett.107.178701`.

Quentin Duchemin and Yohann De Castro. Random geometric graph : Some recent developments and perspectives. *High Dimensional Probability IX : The Ethereal Volume*, pages 347–392, 2023.

Ronen Eldan, Dan Mikulincer, and Hester Pieters. Community detection and percolation of information in a geometric setting. *Combinatorics, Probability and Computing*, 31(6) :1048–1069, 2022.

Paul Erdős, Alfréd Rényi, et al. On the evolution of random graphs. *Publications of the*, 1960.

Santo Fortunato. Community detection in graphs. *Physics reports*, 486(3-5) : 75–174, 2010.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. The geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Connectivity in random annulus graphs and the geometric block model, 2020. URL `https://arxiv.org/abs/1804.05013`.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Community recovery in the geometric block model, 2023. URL `https://arxiv.org/abs/2206.11303`.

Julia Gaudio and Charlie K. Guan. Sharp exact recovery threshold for two-community euclidean random graphs, 2025. URL `https://arxiv.org/abs/2501.14830`.

Julia Gaudio and Andrew Jin. Exact recovery in the geometric sbm, 2025. URL `https://arxiv.org/abs/2512.22773`.

Julia Gaudio and Andrew Jin. Exact recovery in the geometric hidden community model. *arXiv preprint arXiv :2601.17591*, 2026.

Julia Gaudio, Charlie Guan, Xiaochun Niu, and Ermin Wei. Exact label recovery in euclidean random graphs. *arXiv preprint arXiv :2407.11163*, 2024a.

Julia Gaudio, Xiaochun Niu, and Ermin Wei. Exact community recovery in the geometric sbm, 2024b. URL `https://arxiv.org/abs/2307.11196`.

Edward N Gilbert. Random plane networks. *Journal of the society for industrial and applied mathematics*, 9(4) :533–543, 1961.

Jean-Sébastien Gomez, Aurélien Vasseur, Anaïs Vergne, Philippe Martins, Laurent Decreusefond, and Wei Chen. A case study on regularity in cellular network deployment. *IEEE wireless communications letters*, 4(4) :421–424, 2015.

Stefano Guarino, Enrico Mastrostefano, and Davide Torre. Random hyperbolic graphs with arbitrary mesoscale structures. *Physical Review E*, 112(5) :054310, 2025.

Desmond J Higham, Marija Rašajski, and Nataša Pržulj. Fitting a geometric graph to a protein–protein interaction network. *Bioinformatics*, 24(8) :1093–1099, 2008.

Paul W Holland, Kathryn Blackmond Laskey, and Samuel Leinhardt. Stochastic blockmodels : First steps. *Social networks*, 5(2) :109–137, 1983.

Bianka Kovács, Barnabás Benedek, and Gergely Palla. Community detection in hypergraphs through hyperedge percolation. *Scientific Reports*, 15(1), October 2025. ISSN 2045-2322. `https://doi.org/10.1038/s41598-025-19974-9`. URL `http://dx.doi.org/10.1038/s41598-025-19974-9`.

Dmitri Krioukov. Clustering implies geometry in networks. *Physical Review Letters*, 116(20), May 2016. ISSN 1079-7114. `https://doi.org/10.1103/physrevlett.116.208302`. URL `http://dx.doi.org/10.1103/PhysRevLett.116.208302`.

Günter Last and Mathew Penrose. *Lectures on the Poisson process*, volume 7. Cambridge University Press, 2018.

Shuangping Li and Tselil Schramm. Spectral clustering in the gaussian mixture block model, 2024. URL `https://arxiv.org/abs/2305.00979`.

Ye Li, Chaofeng Sha, Xin Huang, and Yanchun Zhang. Community detection in attributed graphs : An embedding approach. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), Apr. 2018.

Siqi Liu, Sidhanth Mohanty, Tselil Schramm, and Elizabeth Yang. Testing thresholds for high-dimensional sparse random geometric graphs. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 672–677, 2022.

Cheng Mao, Yihong Wu, and Jiaming Xu. Random geometric graphs with smooth kernels : sharp detection threshold and a spectral conjecture, 2026. URL `https://arxiv.org/abs/2602.14998`.

Pierre-Loïc Méliot. Asymptotic representation theory and the spectrum of a random geometric graph on a compact lie group. 2019.

Elchanan Mossel, Joe Neeman, and Allan Sly. Stochastic block models and reconstruction, 2012. URL `https://arxiv.org/abs/1202.1499`.

Sandrine Péché and Vianney Perchet. Robustness of community detection to random geometric perturbations. *Advances in Neural Information Processing Systems*, 33 :17827–17837, 2020.

Mathew Penrose. *Random geometric graphs*, volume 5. OUP Oxford, 2003.

Mathew D. Penrose. Connectivity of soft random geometric graphs. *The Annals of Applied Probability*, 26(2), April 2016. ISSN 1050-5164. `https://doi.org/10.1214/15-aap1110`. URL `http://dx.doi.org/10.1214/15-AAP1110`.

Carl Rasmussen. The infinite gaussian mixture model. In S. Solla, T. Leen, and K. Müller, editors, *Advances in Neural Information Processing Systems*, volume 12. MIT Press, 1999. URL `https://proceedings.neurips.cc/paper_files/paper/1999/file/97d98119037c5b8a9663cb21fb8ebf47-Paper.pdf`.

Douglas A Reynolds et al. Gaussian mixture models. *Encyclopedia of biometrics*, 741(659-663) :3, 2009.

Nicolò Ruggeri, Martina Contisciani, Federico Battiston, and Caterina De Bacco. Community detection in large hypergraphs. *Science Advances*, 9(28), July 2023. ISSN 2375-2548. `https://doi.org/10.1126/sciadv.adg9159`. URL `http://dx.doi.org/10.1126/sciadv.adg9159`.

Abishek Sankararaman and François Baccelli. Community detection on euclidean random graphs. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 2181–2200. SIAM, 2018.

Abishek Sankararaman, Haris Vikalo, and François Baccelli. Comhapdet : a spatial community detection algorithm for haplotype assembly. *BMC genomics*, 21(Suppl 9) :586, 2020.

Riitta Toivonen, Jukka-Pekka Onnela, Jari Saramäki, Jörkki Hyvönen, and Kimmo Kaski. A model for social networks. *Physica A : Statistical Mechanics and its Applications*, 371(2) :851–860, November 2006. ISSN 0378-4371. `https://doi.org/10.1016/j.physa.2006.03.050`. URL `http://dx.doi.org/10.1016/j.physa.2006.03.050`.

Ernesto Araya Valdivia and Yohann De Castro. Latent distance estimation for random geometric graphs. In *NeurIPS*, pages 8721–8731, 2019.